# Machine Learning: Question Bank

Modules I to III: Fundamentals, Supervised Classification, Supervised Regression.

# Part A: Short Answer Questions

*20 questions, 2 to 3 marks each.*

### Module I: Fundamentals

**1.** Define machine learning and give one everyday example.

**2.** What is the difference between a feature and a label?

**3.** Name the three types of machine learning and give one example of each.

**4.** List the five stages of the machine learning pipeline.

**5.** What is the difference between classification and regression? Give one example of each.

**6.** Why do we scale features? Name two scaling methods.

**7.** When do you use one-hot encoding instead of label encoding?

**8.** Name three ways to handle missing values.

### Module II: Classification

**9.** How does KNN classify a new data point? Why is feature scaling important for it?

**10.** What is Gini impurity? Calculate it for a node with 5 Pass and 5 Fail students.

**11.** What does CountVectorizer do? Why is Naive Bayes called "naive"?

**12.** Define support vectors and margin in SVM.

**13.** What is the difference between bagging (Random Forest) and boosting (Gradient Boosting)?

**14.** Define precision and recall. Write the formula for each.

**15.** What does a ROC-AUC value of 0.5 indicate?

**16.** Differentiate between overfitting and underfitting.

**17.** What is the difference between a parameter and a hyperparameter?

**18.** How is Stratified K-Fold different from normal K-Fold?

### Module III: Regression

**19.** Write the equation of simple linear regression and explain the slope and intercept.

**20.** What is the role of the learning rate in gradient descent? Which regularisation technique can reduce coefficients exactly to zero?

# Part B: Long Answer Questions

*20 questions, 8 to 10 marks each.*

### Module I: Fundamentals

**1.** Explain the types of machine learning (supervised, unsupervised and reinforcement) in detail with examples and a comparison table.

**2.** Describe the machine learning pipeline step by step, using a suitable example such as predicting student results.

**3.** Explain the role of NumPy, Pandas and Scikit-learn in machine learning, with a small code example for each.

**4.** Explain data preprocessing techniques: handling missing values, feature scaling (with formulas) and encoding categorical data, using examples.

### Module II: Classification

**5.** Explain the K-Nearest Neighbors algorithm with an example. Discuss the effect of choosing K and the importance of scaling.

**6.** Explain how a Decision Tree is built. Discuss Gini impurity and entropy with a worked example, and explain how overfitting is controlled.

**7.** Explain the Naive Bayes classifier using Bayes' theorem. Show how CountVectorizer converts text into features, and build a spam-classification example.

**8.** Explain Support Vector Machines, covering hyperplane, margin, support vectors, kernels and the C parameter.

**9.** Compare Random Forest and Gradient Boosting. Explain how each works and when you would use each.

**10.** A classifier gives TP = 45, FP = 15, FN = 10, TN = 130. Draw the confusion matrix and calculate accuracy, precision, recall and F1-score. Explain which metric matters in which situation.

**11.** Explain the ROC curve and AUC. How is it different from accuracy as a way of judging a classifier?

**12.** What is an imbalanced dataset? Explain why accuracy is misleading and describe at least four techniques to handle it.

**13.** Explain multi-class classification strategies (One-vs-Rest and One-vs-One). How many models does each need for 5 classes?

**14.** Explain the bias-variance trade-off with the concepts of overfitting and underfitting. How can each problem be detected and fixed?

**15.** Explain cross-validation, including K-Fold and Stratified K-Fold. Compare Grid Search CV and Random Search CV with code.

## Module III: Regression

**16.** Explain simple linear regression with the least squares method. For hours studied x = 1, 2, 3, 4, 5 and marks y = 35, 45, 50, 60, 70, find the regression line and predict the marks for 6 hours.

**17.** Explain multiple linear regression and polynomial regression. When is each used, and how can the degree of a polynomial be chosen?

**18.** Explain gradient descent with its update rule and a worked example. Discuss the learning rate and the batch, stochastic and mini-batch variants.

**19.** List and explain the assumptions of linear regression, and explain how each can be checked.

**20.** Explain regularisation. Compare Ridge (L2), Lasso (L1) and Elastic Net with respect to the penalty, the effect on coefficients and feature selection.